

**Ben Thomas Kilby**

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## **PERSONAL PROFILE**

Computer Science for Games Programming BSc student entering my third year, with strong practical experience across software development, game programming, data analysis, cloud infrastructure, AI, XR development and software engineering. I have approximately seven years of programming experience and five years of game-development experience, progressing from self-directed development into university and professional environments.

I enjoy solving complex technical problems and improving the efficiency, usability and reliability of software. My experience includes developing applications in **C#, Python and C++**, working with **Unity, OpenGL, Azure, CI/CD, databases and data-analysis tools**, and contributing to Agile group projects using structured development practices.

I am particularly interested in **software engineering, solutions architecture, data/AI, cloud technologies and commercial application development**. I approach problems methodically, actively seek feedback and enjoy understanding how individual technical components can be designed and integrated into larger systems.

## **TECHNICAL SKILLS**

**Programming:** C#, Python, C++, JavaScript, SQL

**Game Development:** Unity, game systems, physics, 3D programming, game AI, optimisation

**Graphics:** OpenGL, matrices, indices, buffers, scene graphs, 3D environments

**XR:** Mixed Reality (MR), Virtual Reality (VR), Oculus Quest 3, hand tracking, body locomotion, VR controllers

**Cloud & Infrastructure:** Microsoft Azure, Virtual Machines, web hosting, CI/CD

**Data & AI:** Data analysis, regression, neural networks, benchmarking, Excel, Microsoft Access, SQLite

**Software Engineering:** Agile, Scrum-style teamwork, debugging, CRUD, testing, input sanitisation and data validation, system design and architecture

**Project Management:** Gantt charts, Trello, risk assessment, milestone planning and requirements analysis

**Tools:** Git/version control, Unity, Blender, Microsoft Office, Notion

## **UNIVERSITY EXPERIENCE**

**BSc Computer Science for Games Programming**

## **University of Hull | 2024-Present**

### **Entering Year 3**

Throughout my degree I have worked on a broad range of individual and group software projects covering game development, software engineering, AI, graphics, cloud infrastructure and XR.

### **Software Engineering & Agile Development**

- Worked within Agile development teams to design, develop and deliver software projects to defined requirements and deadlines.
- Collaborated closely with team members, contributing technical ideas, implementation work, presentations and project planning.
- Regularly reviewed and debugged interconnected systems, identifying issues and considering alternative technical approaches.
- Developed an understanding of software architecture, maintainability, modularity and the importance of designing systems that can be extended and integrated efficiently.

### **XR Development (Mixed Reality & Virtual Reality)**

Developed XR applications in **Unity**, working with an **Oculus Quest 3** to explore interactive retail environments.

#### **Mixed Reality project:**

- Developed a retail environment overlaid onto a real classroom environment.
- Enables users to move naturally around the environment using body locomotion and interact with virtual objects using their hands.
- Implemented interactions including **grabbing, inspecting and throwing objects**.
- Considered usability and spatial interaction when designing the experience so users could move through the environment seamlessly.

#### **Virtual Reality project:**

- Developed a retail shopping/shelf-placement game using VR controllers.
- Developed retail environment using royalty free assets.

### **Artificial Intelligence**

Completed university AI coursework covering machine-learning techniques including:

- Linear regression
- Polynomial regression degree 4
- Neural networks
- Model evaluation and benchmarking

Implemented and benchmarked different approaches to determine which model provided the most suitable performance for the given requirements. This involved comparing results and using evidence to inform technical decisions rather than selecting an approach solely on assumption.

## **OpenGL 3D Graphics**

Developed a 3D environment using **C# and OpenGL**, gaining practical experience with lower-level graphics programming.

The project involved working with:

- Transformation matrices
- Indices
- Graphics buffers
- Scene graphs
- 3D environments and rendering pipelines

This strengthened my understanding of how higher-level game engines abstract underlying graphics systems.

## **PROFESSIONAL & UNIVERSITY ROLES**

### **Feedback Engineer**

#### **University of Hull**

- Reviewed and marked student work against established requirements and assessment criteria.
- Supported students working with their own codebases, helping them understand and resolve programming problems.
- Taught practical programming techniques, shortcuts and approaches that could improve development efficiency.
- Explained technical concepts in an approachable manner and adapted support to individual students' levels of understanding.
- Used my own software-development experience to identify problems and provide constructive feedback.

### **Solutions Architecture / Business Intelligence Experience**

#### **Flexworks**

- Developed business-intelligence and data-analysis software in a professional environment.
- Worked with data and software solutions designed to support analysis and business decision-making.

- I worked with manipulating and scraping large datasets and producing / maintaining outputted data to a database.
- Developed an understanding of translating technical capabilities into practical business solutions.
- Due to the commercial nature of the work, further details of the software and its implementation are confidential.

## **Professional Networking & Business Development**

I have attended and participated in a range of business and networking events, including:

- Humber Chamber Expo
- Business Week
- Speed networking events
- Other local professional networking opportunities

These experiences strengthened my confidence in communicating with professionals, discussing technical work and understanding how technology is applied within businesses.

## **GAME DEVELOPMENT & PROJECT EXPERIENCE**

### **Independent Game Development**

Approximately **seven years of game-development experience**, primarily using **Unity and C#**, with three major games developed alongside numerous smaller projects and game jams.

Experience includes:

- Designing and programming reusable game systems.
- Implementing physics using vectors and quaternions.
- Developing modular systems that can be integrated with other components.
- Debugging complex interactions between interconnected systems.
- Optimising systems to reduce unnecessary processing and repetitive code.
- Designing mechanics based on playtester feedback.
- Iterating on projects based on testing and user feedback.
- Planning projects using milestones, Gantt charts, Trello and risk assessments.

### **Game Jams**

Participated in multiple game jams, achieving:

- **1st place** — 3 Thing Game
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- **3rd place** — Arcademia

These projects required rapid development, prioritisation, teamwork and delivering functional software within strict time constraints. For one 3 Thing Game Jam I worked and directed a team of 2 but have mostly taken part on my own.

## **EARLIER IT & DATA EXPERIENCE**

### **Data Analysis & Excel**

Approximately eight months of practical experience using Excel for data entry, analysis and visualisation.

- Created structured data-entry forms.
- Produced graphs to identify and communicate trends.
- Designed dashboards to present data clearly.
- Converted raw data into readable information that could support decision-making.

### **Microsoft Access**

Used Microsoft Access during college coursework to develop relational databases.

- Created relational database structures.
- Filtered and cleaned datasets.
- Produced forms and reports to present information.
- Developed an understanding of relational data modelling and database design.

## **EDUCATION**

### **University of Hull**

#### **BSc Computer Science for Games Programming**

2024–Present | Entering Year 3

### **Wilberforce College**

#### **Level 3 BTEC Extended Diploma in IT — DDD**

2022–2024

Relevant areas included:

- Programming
- Computer Game Development
- IT Project Development
- Data Modelling
- Cybersecurity
- Website Development
- Mobile App Development

- Digital 2D & 3D Graphics
- IT Service Delivery
- Enterprise in IT
- Social Media in Business

## **Malet Lambert School**

**2017–2022**

GCSE results included:

- Computer Science — Grade 8
- English — Grade 7
- Geography — Grade 6
- Mathematics — Grade 5 (foundation)
- IT BTEC Standard Certificate — D\*

## **COURSES & CERTIFICATIONS**

- Udemy — Complete C# Unity Game Developer 3D
- Udemy — Complete Blender Creator: Learn 3D Modelling for Beginners
- CPD Certified — Cyber Security
- CPD Certified — Health & Safety: Risk Assessment
- CPD Certified — Data Protection in Marketing
- CPD Certified — Data Protection & GDPR
- CPD Certified — Information Security

Available upon request.